iEP SkySAFE 2.0

Aviation Safety Assessment Report

Application Reference: APP-002

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IFC filename	ZHA-B-BWK-C-MR-R18.ifc
Report generated	2026-05-19 08:18 UTC
Pipeline version	0.1.0

STAGE 1 -- IFC INGESTION & VALIDATION — PASS

PASS File located on SFTP landing directory

PASS IFC schema validated: IFC4 / IFC+SG v1.0

PASS Filename: ZHA-B-BWK-C-MR-R18.ifc

STAGE 2 -- EXTERIOR SHELL EXTRACTION — PASS

Status	OK
Exterior elements extracted	1,567
Mesh vertices	51,562
Mesh triangles	97,124

STAGE 3 -- HEIGHT ANALYSIS — COMPLETE

Site elevation (IfcSite.RefElevation)	0.0 m (local datum)
Lowest point -- ground slab soffit	-2.05 m
Highest point -- roof apex (local)	69.86 m
Building height	71.91 m
Peak height above mean sea level (AMSL)	**69.86 m AMSL** -- used for all aviation assessments

STAGE 4 -- GROSS FLOOR AREA (GFA)

Proposed GFA	74,749.9 m²

STAGE 4b -- FACADE MATERIAL SURVEY

Unique material types identified: **5**

Material	Elements
NLRS_f2_beton prefab	915
NLRS_f2_beton ihwg	588
NLRS_n7_isolatie EPS	40
NLRS_f1_kalkzandsteen wit	8
Analytical Floor Surface	3

STAGE 5 -- COMPOSITE HEIGHT TEMPLATE (CHT) INTERSECTION

PENETRATION DETECTED

Max permitted height (CHT)	88.85 m AMSL
Building peak height	101.55 m AMSL
Penetration depth	**12.7 m**

Penetrated zones:

Zone	Permitted (m AMSL)	Building (m AMSL)	Excess (m)
CHT Zone A -- Primary Obstacle Limitation	88.85	101.55	12.7

Action required: CHT penetration triggers mandatory ADP officer review. CAAS must assess whether a height reduction, LOC conditions, or a formal aeronautical study is required before a Letter of Consent can be issued.

STAGE 6 -- OLS (OBSTACLE LIMITATION SURFACES) INTERSECTION

1 surface(s) penetrated

Penetrated surfaces:

Surface	Runway	OLS Height (m AMSL)	Building (m AMSL)	Depth (m)
Transitional Surface	RWY 02L/20R	94.13	101.55	7.42

Surfaces clear:

PASS Approach Surface

PASS Take-off Climb Surface

PASS Inner Horizontal Surface

PASS Conical Surface

PASS Outer Horizontal Surface

Note: OLS penetration must be declared in the LOC application and assessed by CAAS aerodromes engineers.

STAGE 7 -- ILS TECHNICAL TEMPLATE IMPACT

1 template(s) breached

Breached templates:

Template	System	Runway	Ceiling (m AMSL)	Building (m AMSL)	Excess (m)
ILS-WSSS-02L-LOC	Localiser (LOC)	RWY 02L	85.0	101.55	16.55

Templates clear:

PASS ILS-WSSS-02L-GP

PASS ILS-WSSS-20R-LOC

PASS ILS-WSSS-20R-GP

Action required: ILS template breach requires an RF propagation study by the CAAS ILS team before the LOC application can progress.

ASSESSMENT SUMMARY

Check	Result	Detail
IFC Ingestion & Validation	PASS	ZHA-B-BWK-C-MR-R18.ifc

Exterior Shell Extraction	**PASS**	1,567 elements, 97,124 triangles
Height Analysis	**COMPLETE**	71.91 m building height, 69.86 m AMSL peak
Gross Floor Area	*PENDING*	74,749.9 m^2
Composite Height Template (CHT)	**FAIL**	Penetration: 12.7 m
OLS Intersection	**FAIL**	1 surface(s) penetrated
ILS Technical Template	**FAIL**	1 template(s) breached